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Git Repo/ Colab Link: https://github.com/divinagraciabelgica-wq/Belgica_Elec2 Date: 03.15.26

Source: departuredelays.csv

This dataset contains domestic U.S. flight departure delay records. It includes information about flight dates, delay duration in minutes, flight distances, and airport codes for origin and destination.

Total records: 1,391,578 | Missing values: None

```
============================================================
SECTION 1: Loading Datasets
============================================================
✅ Delays  : (1391578, 5)
✅ Housing : (20640, 10)
✅ Airbnb  : (7151, 106)
✅ Covid   : (429435, 67)

============================================================
SECTION 2: Data Preprocessing
============================================================

[Delays] Missing values:
date           0
delay          0
distance       0
origin         0
destination    0
dtype: int64
```

Source: housing.csv

This dataset contains housing data from the 1990 California census. Each row represents a block group (a small geographic area) and includes information about housing age, number of rooms, income levels, house values, and proximity to the ocean

Total records: 20,640 | Missing values: 207 (total_bedrooms only)

```
[Housing] Missing values:
longitude               0
latitude                0
housing_median_age      0
total_rooms             0
total_bedrooms        207
population              0
households              0
median_income           0
median_house_value      0
ocean_proximity         0
dtype: int64
```

Source: sf-airbnb.csv

This dataset contains San Francisco Airbnb listing data scraped in March 2019. It has 106 columns covering property details, host information, location, pricing, availability, and guest review scores.

Total records: 7,151 | Missing values: 1,421 (review_scores_rating)

```
[Airbnb] Missing values (selected columns):
property_type                0
room_type                    0
neighbourhood_cleansed       0
price                        0
bedrooms                     2
bathrooms                   21
review_scores_rating      1421
dtype: int64
```

Source: owid-covid-data.csv

This dataset contains global COVID-19 statistics compiled by Our World in Data. It tracks daily case counts, deaths, vaccinations, and other health indicators across 67 columns for countries and regions worldwide.

```
[Covid] Missing values (selected columns):
continent             26525
location                  0
date                      0
total_cases           17631
new_cases             19276
total_deaths          17631
people_vaccinated    348303
dtype: int64
```

Total records: 429,435 | Missing values: Significant (especially people_vaccinated: 348,303)

Drop Missing Rows: dropna() was applied after selecting relevant columns. Row counts before and after.

```
[Delays]  Before: 1,391,578  | After: 1,391,578
[Housing] Before: 20,640 | After: 20,433
[Airbnb]  Before: 7,151  | After: 5,708
[Covid]   Before: 429,435 | After: 60,745

[Delays] Selected columns:
       date  delay  distance origin destination
0  1011245      6       602    ABE         ATL
1  1020600     -8       369    ABE         DTW
2  1021245     -2       602    ABE         ATL
```

Select Columns: Only the most relevant columns were kept for each dataset to reduce memory and improve clarity.

```
[Housing] Selected columns:
   housing_median_age  median_income  median_house_value ocean_proximity
0                41.0         8.3252           452600.0        NEAR BAY
1                21.0         8.3014           358500.0        NEAR BAY
2                52.0         7.2574           352100.0        NEAR BAY

[Airbnb] Selected columns:
  property_type        room_type neighbourhood_cleansed    price  bedrooms  \
0     Apartment  Entire home/apt       Western Addition  $170.00       1.0
1     Apartment  Entire home/apt         Bernal Heights  $235.00       2.0
2     Apartment     Private room         Haight Ashbury   $65.00       1.0

   bathrooms  review_scores_rating
0        1.0                  97.0
1        1.0                  98.0
2        4.0                  85.0

[Covid] Selected columns:
    continent     location        date  total_cases  new_cases  total_deaths  \
414      Asia  Afghanistan  2021-02-22      55604.0        0.0        2432.0
420      Asia  Afghanistan  2021-02-28      55714.0      110.0        2443.0
436      Asia  Afghanistan  2021-03-16      55985.0        0.0        2457.0

     people_vaccinated
414                0.0
420             8200.0
436            54000.0
```

Filtering: Conditional filters were applied to extract meaningful subsets

```
[Delays] Flights with delay > 0: 591,727
[Housing] Houses above $200,000: 8,621
[Airbnb] Listings rated >= 90: 5,145
[Covid] Records with new_cases > 0: 8,707
```

Basic Statistics: describe() was used to generate statistical summaries for key numeric columns.

```
[Delays] Delay stats:
count    1.391578e+06
mean     1.207980e+01
std      3.880773e+01
min     -1.120000e+02
25%     -4.000000e+00
50%      0.000000e+00
75%      1.200000e+01
max      1.642000e+03
Name: delay, dtype: float64

[Housing] House value stats:
count     20433.000000
mean     206864.413155
std      115435.667099
min       14999.000000
25%      119500.000000
50%      179700.000000
75%      264700.000000
max      500001.000000
Name: median_house_value, dtype: float64
```

```
[Airbnb] Review score stats:
count    5708.000000
mean       95.557989
std         6.940153
min        20.000000
25%        94.000000
50%        98.000000
75%       100.000000
max       100.000000
Name: review_scores_rating, dtype: float64

[Covid] Total cases stats:
count    6.074500e+04
mean     4.707989e+06
std      1.161739e+07
min      0.000000e+00
25%      1.281420e+05
50%      7.433430e+05
75%      3.985606e+06
max      1.033436e+08
Name: total_cases, dtype: float64
```

Writing to External Storage: Query results were saved to three different file formats inside a /content/output/ directory:

```
============================================================
SECTION 4: Writing Results to External Storage
============================================================
✅ CSV files saved to /content/output/
✅ JSON files saved to /content/output/
✅ TXT report saved to /content/output/summary_report.txt
```

Top 5 Most Frequent Origin Airports Insight: Atlanta (ATL) is the busiest origin airport with over 91,000 flights. All top 5 are major hub airports in the US, which aligns with real-world air traffic patterns.

```
[Query 1] Top 5 Most Frequent Origin Airports:
origin  flight_count
   ATL         91484
   DFW         68482
   ORD         64228
   LAX         54086
   DEN         53148
```

Top 5 Ocean Proximity Categories (Housing) Insight: Most California housing blocks are within 1 hour of the ocean, reflecting California's coastal population concentration. Island properties are extremely rare with only 5 records.

```
[Query 2] Top 5 Ocean Proximity Categories (Housing):
ocean_proximity  count
      <1H OCEAN   9034
         INLAND   6496
     NEAR OCEAN   2628
       NEAR BAY   2270
         ISLAND      5
```

Top 5 Airbnb Property Types Insight: Apartments dominate the SF Airbnb market, reflecting the city's dense urban character. Houses are second most common.

```
[Query 3] Top 5 Most Frequent Airbnb Property Types:
 property_type  count
     Apartment   2179
         House   1736
   Condominium    613
   Guest suite    464
Boutique hotel    139
```

Top 5 Locations by Total COVID-19 Cases Insight: The United States recorded the highest total COVID-19 cases. The top 5 countries represent a mix of both Asian and European nations with large populations and high testing rates.

```
[Query 5] Top 5 Locations by Total COVID Cases:
     location  max_total_cases
United States      103343569.0
        China       98637553.0
        India       45041748.0
       France       38997490.0
      Germany       38356872.0
```

Top 5 Origins with Highest Average Delay Insight: Smaller regional airports have the highest average delays. This is likely due to fewer resources and limited recovery options when delays occur compared to major hubs.

```
[Query 6] Top 5 Origins with Highest Average Delay (min):
origin  avg_delay
   SWF      74.98
   ALO      73.74
   GUM      71.89
   BGR      70.94
   CEC      68.46
```